Grade 6 Accelerated Unit 9 Assessment Guide

General Information	Unit 9 introduces students to finding and interpreting measures of center and measures of variability. Students also learn how to read box plots and histograms in preparation for the students actually creating the box plots or histograms. To help teachers determine if students struggle distinguishing between the mean and median, these measures of center were not asked within the same context. Students begin to collect data and organize the data using representations in kindergarten. In grade 5, students begin to find statistical values. The benchmarks on this assessment include focus on creating and interpreting displays. Students first saw line plots (dot plots) and bar graphs in grade 3. It is helpful for a 6th grade accelerated teacher to know what types of statistical displays students have experienced in kindergarten through fifth grade. Having this information may help teachers bridge gaps that students may have in creating a box plot or histogram. The benchmarks in this unit also have students using line plots (dot plots). Students will use line plots (dot plots) and box plots in Unit 10 and Algebra 1 as well. The wording of the five benchmarks in this unit is important to note. For example, MA.6.DP.1.3 and MA.6.DP.1.4 begin with “given a...”, so for these benchmarks the student will not be creating a statistical display, but the statistical display should be given. Because creating displays can take time, this is an important distinction. Students are allowed the use of a calculator for this assessment because the focus should be on understanding how to find each measure. Most of the questions on this assessment are not calculator dependent. Teachers should note that an assessment limit in the draft item specifications states that items that assess finding the mean are restricted to 10 values. This restriction is likely in place so that difficulty is easier so that students could do the mathematics without a calculator. Teachers could easily reorder the questions to create a two-part assessment where one part is calculator and one part is not calculator, if preferred.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.6.DP.1.2
38		Recommended Scoring (5 Total Points)	Consider not giving partial credit.
Question	**1b**	**Benchmark(s)**	MA.6.DP.1.2
22		Recommended Scoring (5 Total Points)	Consider not giving partial credit.
Question	**1c**	**Benchmark(s)**	MA.6.DP.1.2
30		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	2	Benchmark(s)	MA.6.DP.1.5
The image part with relationship ID rId5 was not found in the file.		**Recommended Scoring** **(9 Total Points)**	Consider giving students 3 points for each correct label.
Question	**3**	**Benchmark(s)**	MA.6.DP.1.4
☐ 10% of the data are between H and T. ☐ 25% of the data are between T and N. ☐ 25% of the data are between R and H. ☐ 50% of the data are between H and N. ☐ 50% of the data are between R and N. ☐ 75% of the data are between N and B.		**Recommended Scoring** **(15 Total Points)**	Consider giving students 4 points for each correct value. Consider a deduction of 5 points for each incorrect selection.

Question	4	Benchmark(s)	MA.6.DP.1.1
The image part with relationship ID rId5 was not found in the file. *Student could also correctly complete the statement saying that Erika did not write a statistical question.*		**Recommended Scoring** **(3 Total Points)**	Consider not giving partial credit since students should answer all parts of the statement correctly to show understanding.

Question	5	Benchmark(s)	MA.6.DP.1.5
The image part with relationship ID rId5 was not found in the file.		**Recommended Scoring** **(15 Total Points)**	Consider giving students 5 points for each correct value. Consider a deduction of 5 points for each incorrect selection.

Question	**6**	**Benchmark(s)**	MA.6.DP.1.5
The image part with relationship ID rId5 was not found in the file.		**Recommended Scoring** **(15 Total Points)**	Consider giving students 3 points for each correctly plotted value in the five-number summary.
Question	**7**	**Benchmark(s)**	MA.6.DP.1.1
A. What is the cost, in dollars, of a healthy meal? B. How many countries have a healthy meal that costs \$3.50? C. Which countries have healthy meals that cost less than \$5.50? D. How much, per person, does it cost to make a meal in the United States?		**Recommended Scoring** **(3 Total Points)**	Partial credit is not recommended.

Question	8a	Benchmark(s)	MA.6.DP.1.4
The image part with relationship ID rId5 was not found in the file.		**Recommended Scoring** **(20 Total Points)**	Consider giving students 5 points for each correctly completed answer box. Each answer box represents a slightly different concept.
Question	**8b**	**Benchmark(s)**	MA.6.DP.1.2
17.5		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.